Chains of Chance: A Convoluted History of Pattern Recognition

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Picture Question

When black grids appear out of the void and lines suddenly change direction, chance is at work on a composition. Consider two pictures that could not be more different and yet at some level are also very similar: Both contain four squares with different patterns, but in the first one, highly varied angles fill the surfaces of the squares (fig. 1, left), while in the second, the surfaces are filled with rectangular pixels in varying shades of gray (see fig. 1, right). The first is an artwork by Frieder Nake, the second a "plate" by Benoît Mandelbrot illustrating a scientific model. What constitutes the essential identity of these pictures?

In the Face of the Irregular

It is well known that Benoît Mandelbrot begins the book *The Fractal Geometry of Nature* with a declaration: "There is a fractal face to the geometry of nature."[1] In this "scientific essay," he discloses that many forms of nature obey one and the same principle, namely self-similarity or, to use Mandelbrot's term, *scale invariance*. Based on this principle—according to which the infinite enlargement of a patterned object continuously reveals the original pattern repeated at ever greater scales—fractal geometry reveals its "hidden face."[2] The following is an attempt to retrace the path to fractals, which itself resembles the

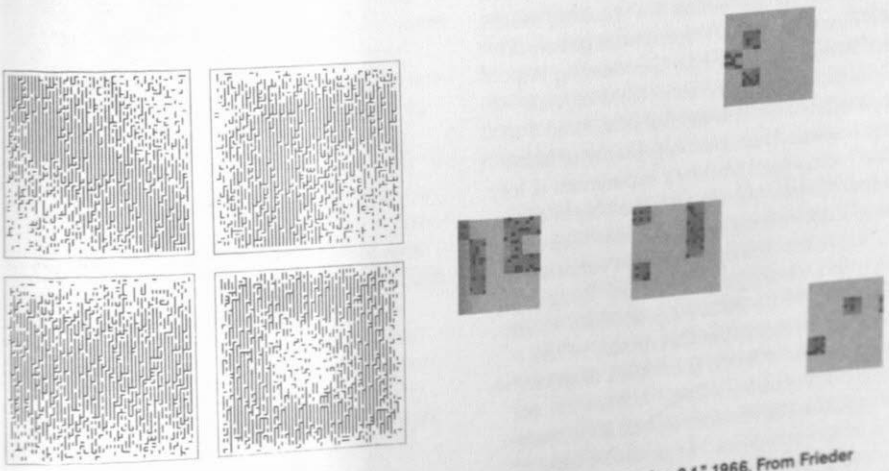


Fig. 1. (left) Frieder Nake, "Four Realizations of Walk-Through Rasters, from Series 2.1," 1966. From Frieder Nake, *Ästhetik als Informationsverarbeitung. Grundlagen und Anwendungen der Informatik im Bereich ästhetischer Produktion und Kritik* (Vienna: Springer, 1974), 236, illus. 5.5-5. With kind permission of Springer Science and Business Media. (right) Benoît Mandelbrot, "Implementation of Hoyle's Model (Dimension D=1) Using Random Curdling in a Grid," computer drawing. From Benoît Mandelbrot, *The Fractal Geometry of Nature* (New York: Freeman, 1982), 219.

curves analyzed in Mandelbrot's book. He cryptically alludes to the path in the book's epilogue, where he speaks of a land (which he does not name) with which he has had a lifetime involvement; of a review that he retrieved from the wastebasket of a "pure" mathematician; of regularities in word frequencies and the birth of mathematical linguistics; of accidents, probabilities, and chains—and of the fact that though all of these things belong to the past, he has not stopped sensing their aftereffects.[3] The path consisted of the use of certain conjectural techniques to derive order from chance processes. It was out of this that the "geometric face" of the theory of scaling emerged, that it "grew in importance and gave size to fractal geometry."[4]

The Instrument of Chance

On January 23, 1913, the Russian mathematician Andrei Andreyevich Markov (1856–1922) presented a meeting of the Physical-Mathematical Division of St. Petersburg's Academy of Sciences with the results of an experiment he had conducted on the letters of the Russian alphabet.[5] His goal was to establish the concept of a chain of chance variables, and the resulting article, "An Example of Statistical Investigation of the Text *Eugene Onegin* Concerning the Connection of Samples in Chains," was published a short time later in Volume 7 of the *Isvestia Imperatorskoj Akademii Nauk*. The radicality of Markov's experiment is fully apparent from his report.[6] The object of his statistical investigation had been *Eugene Onegin*, the Romantic novel in verse by the Russian literary giant Alexander Sergeyevich Pushkin (1799–1837) (fig. 2). Fully aware that it was the 200th anniversary of the publication of Jacob Bernoulli's classic treatise on probability theory,[7] Markov set out to analyze the Russian classic as a system of *linked quantities*. His gridlike blocks of sharply printed letters became the object of his theoretical investigation. Disregarding all punctuation marks and spaces between words, he arranged the first 20,000 letters from *Eugene Onegin* into 200 blocks of

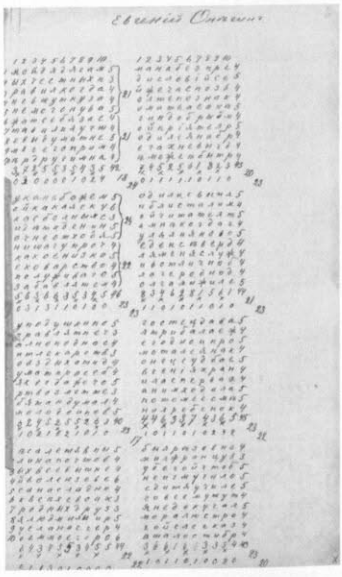


Fig. 2. Andrei A. Markov. "Pushkin in Block Form." From *Calculations on Eugene Onegin*, 1913. © St. Petersburg Branch of the Archive of the Russian Academy of Sciences, 2012. Holding 173, Register 1, File 118, Sheet 2.

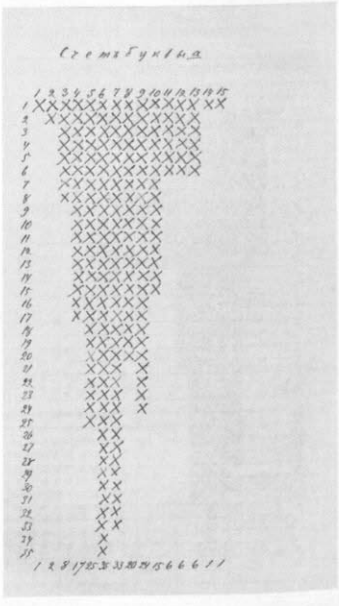


Fig. 3. Andrei A. Markov. "The Altered Count of the Letter A". From *Calculations on Eugene Onegin*, 1913. © St. Petersburg Branch of the Archive of the Russian Academy of Sciences, 2012. Holding 173, Register 1, File 118, Sheet 12r.

letters, each with ten columns and ten rows. From the interpretive section of Markov's report, it is clear that he wanted to produce a model from which one might derive the probabilities of the alternation between vowels and consonants that is typical of Russian as well as English and many other languages (fig. 3). He felt that it should be possible to predict the appearance of a given letter from the letters that had preceded it. In this way, language was transformed into an arrangement of links on paper that served as the basis of a new probability theory.

The Biometry of Literary Style

Markov's study did not wait long for a response. In 1915, in Volume 20 of the St. Petersburg *Isvestia Akademii Nauk, otdelenie russkogo jazyka* (section of Russian language), Nikolai Aleksandrovich Morozov (1854–1946) published an analysis of literary style. Morozov's "linguistic spectra"—the phrase appears in the title of his article—were intended to serve as "a way to distinguish plagiarisms from the genuine works of a given author," and their sophisticated linkings were to provide objective proof of the linguistic pattern system.[8] Morozov, a poet, writer, and linguist who would become an honorary member of St. Petersburg's Academy of Sciences, had acquired his scientific knowledge through lay experiments in cosmology, physics, chemistry, biology, geophysics, and meteorology, which had taught him that true accuracy in science is dependent entirely on statistics.[9] Morozov now wanted to use statistics as the basis for a precise, scientific analysis of language. He wrote: "If in nature and everyday life all manner of events appear at first to be coincidental, but after a certain number of repetitions take on a certain predictability, why cannot the same be true of language? The repetitions of words in our oral and written language must also demonstrate various probabilities."[10] The regularities of nature that Morozov had observed in heavenly bodies, mechanical processes, and the composition of matter he also saw in the way every author uses particular turns of phrase:[11] "Some writers frequently use the word 'which,' others don't like it, and replace it with the participial form of the verb, which is added at the end. Some use the particle 'between,' others write in place of it: 'amid' or 'within.'"[12]

Morozov postulated that all such distinctions in the structure of language can be traced back to "purely *mechanical causes*" consisting of "linked chains with internal and external linguistic influences" that "have long since become established in the realm of the unconscious."[13] For precisely this reason, Morozov accorded to particles a "switching function." According to his interpretation, such mechanical processes serve to regulate and organize our speech and the writing of poets.[14]

In order to analyze authorship questions, Morozov needed a method built on "reliable material." He therefore concentrated on linguistic chains, not on conceptual ones, which would "differ with every original writer."[15] Linguistic sequences are made up of individual elements "that in specific speech chains are not only typical of nearly all languages, but also appear in the same way in nearly all literary genres" and therefore can be termed "*functional* or *organizing* particles of human speech."[16] One such particle is the Russian negation "*ne*," which Morozov decided to study in the writing of nineteenth-century authors. To him, questions of authorship became a game of numbers:

> Count, and you will see that in every thousand words in Tolstoy this [negation] appears on average somewhat fewer than twenty times, in Pushkin and Gogol roughly twenty times, and in Turgenev a bit more than in his predecessors, in the realm of between twelve and thirty-five times per thousand words, always depending on the given writer's proclivity to negation.[17]

Morozov thus observed that the use of "the negative particle '*ne*' depends in large part on individual variation," and accordingly can be said to characterize the structure

of each author's language.[18] This appears to contradict the proposition that language patterns can be predicted via the logic of the chain: In Morozov's system, the author's uniqueness is analogous to the uniqueness of the given linguistic work. His implicit criticism of Markov becomes all the more apparent when at the end of his introduction he states: "Unfortunately, you will not be able to arrive at any conclusion if all the words are written together without spaces and without punctuation."[19] Despite this

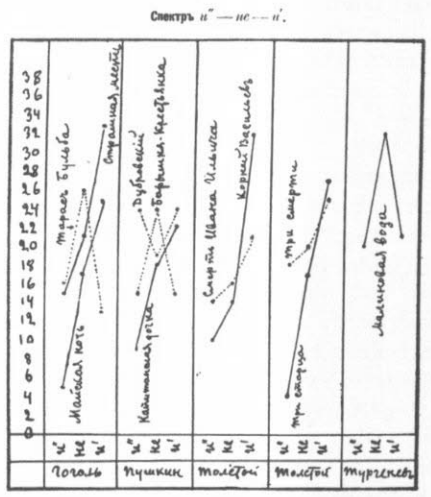


Fig. 4. "Linguistic Spectra." From N. A. Morozov, "Lingvističeskie spektry. Sredstvo dlja otličenija plagiatov ot istinnych proizvedenij togo ili drugogo izvestnogo avtora," *Isv. Akad. Nauk, otd. Russ. Jaz., Sankt Peterburg*, vol. 20 (1915), table IV, 108. © Moscow Branch of the Archive of the Russian Academy of Sciences, 2012.

objection, Morozov was directly indebted to Markov. He not only took up his predecessor's experiment, he, like Markov, was driven by a desire to visualize chance. Indeed, the graphics he produced in his pattern experiments led him to identify authors' textual quirks and to locate them within linguistic spectra (fig. 4). But though Morozov's research was based on pictures of chance, it was published as "linguistic analysis."[20] In order to differentiate his spectra, Morozov broke them down into prepositional, subjunctive, pronominal, and other syntactic groupings and superimposed the corresponding graphs. At this point, his picture took on an authority that seemed to exceed that of the linguistic expert. An overlay of the graphs made it clear that "with only a simple glance at the prepositional spectrum of any given work of Pushkin that was possibly erroneously attributed to Gogol we would have recognized the inaccuracy of that assertion and accordingly expressed the suspicion that the work was likely one of Pushkin's."[21] However, as a statistician, Morozov knew that only averages can be compared in this manner. His overlay graphs could not serve as incontrovertible proof. Even so, he initially concentrated on 1,000 words, from which he calculated a mean for the particles. He identified this mean as a natural value that determined what he called the author's "syllable physiognomy."[22] The spectra were sufficiently precise that they could be said to reveal the author's specific signature and set him apart from other writers. Here "physiognomy" is not merely a figure of speech: The visible countenance of language, represented by an overlay of graphs, became a mark of individuality, one with which it was possible to distinguish genuine authors from false ones. Chance is not without order. When Mandelbrot later felt that he was looking at the "fractal face of nature," he was translating this same experience into his theory and his pictures.

True to the principle of the chain reaction, Markov, the man who initiated the notion, responded to Morozov's provocation a year later. He asserted that Morozov's idea was not fundamentally wrong, but that its implementation should by no means be based on a subjective sense of literary style, and that counting a mere 1,000 words was not enough. He thereby dismissed Morozov's theory of the spectra embodied in the example of the negation as unsubstantiated by either literature or mathematics. Markov, who never tired of counting, used Morozov's own method against him, raising threefold the count of negations in the given authors and thereby demonstrating the inadequacy of Morozov's linguistic spectra as opposed to his own chain patterns. Markov insisted that precise, scientific analysis of language, as undertaken by Morozov, would have to rely on a mathematical-statistical foundation, one that presupposed the counting of several thousand sentences, words, syllables, and letters.[23] Since the invention of the computer, automatic speech recognition has been achieved on just such a basis, its fundamental problem being the establishment of a connection between the sequence of acoustical sounds and the words that correspond to it.

The Principle of Least Effort

The pattern systems of the 1910s, which were based on the frequencies of letters and individual words, were again taken up in the 1930s by George Kingsley Zipf. A lecturer at Harvard, Zipf expanded Markov and Morozov's methods in his own research by investigating the frequency of words in various languages, information that he used to recognize new speech patterns.[24] Zipf arranged the words in a language according to their frequency of appearance, and he managed to derive from this various mathematical constants. For example, the more frequently a word is used, the shorter it is. A word's age also has to do with how often it is used.[25] These laws would eventually find their way into the major study on communication that he published in 1949 under the title *Human Behavior and the Principle of Least Effort*.[26] There Zipf applied his mathematical findings to people's speech habits: Frequently-used words are shortened in their sound structure whereas greater length is tolerated in less common words. It was from this observation that Zipf derived his model of communication: Every speaker wishes to communicate his message with the least possible effort. Yet he encounters a difficulty, for the listener or reader on the other end of the channel wants to decipher that message with the least possible effort.[27] It is only when these two opposing interests are brought into balance that communication is possible.

Among other things, it was a review of this book that led Mandelbrot to fractals: "This empirical regularity concerning word frequencies had come to my attention through a book review. The event seems too symbolic to be true, but the review in question had been retrieved from a 'pure' mathematician's wastebasket, for light reading on the Paris subway."[28] The pure mathematician was Mandelbrot's uncle, Szolem Mandelbrojt, with whom Mandelbrot frequently stayed between 1945 and 1947 while studying at the École polytechnique and whom he visited regularly while working on his doctorate. The reviewer was in fact the mathematician Joseph Leonard Walsh, who in 1949 discussed the study of his Harvard colleague Zipf in *Scientific American*. As chance would have it, in his dissertation, "Contribution à la théorie mathématique des jeux de communication,"[29] Mandelbrot was combining previously known theories of linguistic analysis with Norbert Wiener's cybernetics and John von Neumann's game theory, which were also in vogue at that time. By analyzing language from a macrolinguistic perspective rather than a microlinguistic one, Mandelbrot hoped his study would make him "the Kepler of mathematical linguistics."[30] He placed himself in the Markov tradition but pointed out that the Russian's theory had meanwhile been expanded beyond dependent chance quantities by a series of new concepts and interpretations that had changed the shape of his chains. This further development was mainly owing to Claude E. Shannon, who in 1947 showed that the Markov chain was still useful as a mathematical model for the theory of information transmission. As Shannon's illustration shows, code elements can quantify the information to be communicated with numbers (fig. 5, top). Shannon relied on the theory of Markov chains. "The main point at issue," he wrote, "is the effect of statistical knowledge about the source in reducing the required capacity of the channel, by the use of proper encoding of the information"[31] (see fig. 5, bottom). Mandelbrot, in turn, expanded this model for his own purposes; what was crucial in its arrangement of signs was the introduction of a space sign, one that cannot appear within a word. It was an element that mathemati-

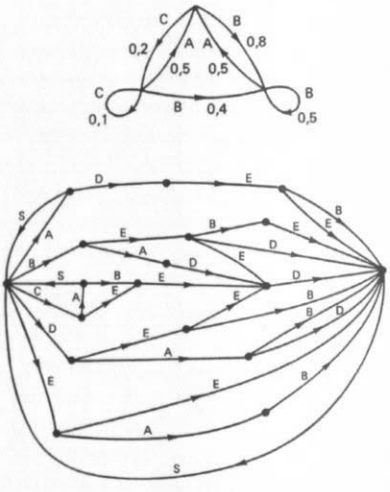


Fig. 5. "Markov Processes." From Claude E. Shannon, *Mathematical Foundations of Information Theory* (New York: Dover, 1976), 16, illus. 4–5.

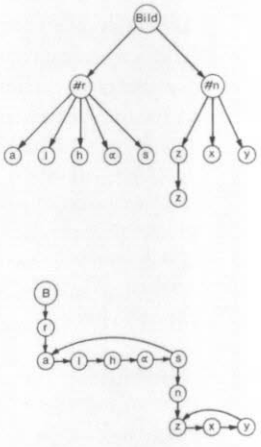


Fig. 6. "Markov Processes." From Frieder Nake, *Ästhetik als Informationsverarbeitung. Grundlagen und Anwendungen der Informatik im Bereich ästhetischer Produktion und Kritik* (Vienna: Springer, 1974), 235, illus. 5.5-5. With kind permission of Springer Science and Buisness Media.

cians like Markov and Shannon had omitted in their language patterns but that linguists like Morozov and Zipf had reintroduced. Merely marking the boundaries between words, the sign is not dependent on adjacent letters.[32] As a result, the individual word becomes the central interval of chance,[33] and from the point of view of probability theory, it is revealed that language shows a high degree of self-similarity. Mandelbrot recognized in this the scale invariance that would become one of the fundamental principles of his later work.

One also sees this in his implementation of the cosmological model of Fred Hoyle, which was based on a chance curdling of forms in a grid (see fig. 1, right). Regardless of where and in which shade of gray the grids appear, they are determined by the probabilities of transition. The same is true of the angles in Frieder Nake's *Walk-Through Raster* series (see fig. 1, left): Here it is a Markov chain of chance events that determines where a given transition will occur (fig. 6).[34] In the case of Mandelbrot's Hoyle visualizations, the chance events are determined by means of a transition probability, be it the flip of a coin, a given birthrate, or the calculated distribution of galaxies that influence degrees of density. The codes and the sources are thus linked together. It was for this reason that Mandelbrot wrote at the end of *The Fractal Geometry of Nature*: "At a critical point in my life I read a wise review of *Human Behavior* by the mathematician J. L. Walsh. By only mentioning what was good, this review influenced greatly my early scientific work, and its indirect influence continues. Therefore, I owe a great debt to Zipf through Walsh."[35] Without the attempts to make visible the linkings of chance that began with Markov, Mandelbrot would not have been able to recognize the "fractal face of nature."

1 Benoît Mandelbrot, *The Fractal Geometry of Nature* (New York: Freeman, 1982), 3.
2 Ibid., 4.
3 Ibid., 422–24.
4 Ibid., 424.
5 For a detailed description of the experiment and its ramifications, see Philipp von Hilgers and Wladimir Velminski, eds., *Andrej A. Markov. Berechenbare Künste. Mathematik, Poesie, Moderne* (Zurich: diaphanes, 2007).
6 Aleksandr Chuprov points to the importance of Markov to his theory. In a posthumous review that discusses Markov's fourth edition of his probability calculation (Markov 1924), Chuprov suggests to his longtime correspondent that his general approach relied on complex avenues, and that Markov "assumes great importance when he is taken up by a broad scientific background, which he deserves." See Hilgers and Velminski, eds., *Andrej A. Markov*, 32–39.
7 Markov was very much involved in the organization of its celebration. See the letter from Markov to Chuprov from January 15, 1913, in A. A. Markov, "Primer statističeskogo issledovanija nad tekstom 'Evgenija Onegina,' illyustrirujuščij svjaz' ispytanij v zep," *Izv. Imp. Akad. Nauk, Sankt Peterburg*, ser. 8, vol. 7, no. 3 (1913), 153–62.
8 N. A. Morozov, "Lingvističeskie spektry. Sredstvo dlja otličenija plagiatov ot istinnych proizvedenij togo ili drugogo izvestnogo avtora," *Isv. Akad. Nauk, otd. Russ. Jaz.*, Sankt Peterburg, vol. 20 (1915), 93–134. See also Wladimir Velminski, "Der Speck am Text. Die Entstehung der Genetik aus der Berechenbarkeit der Literatur," in Hilgers and Velminski, eds., *Andrej A. Markov*.
9 N. A. Morozov, *Povest'moej žizni* (Moscow: Izd-vo Akademii nauk SSSR, 1961), 379.
10 Morozov, "Lingvističeskie spektry," 100.
11 See N. A. Morozov, *Die Evolution der Materie auf den Himmelskörpern: Eine theoretische Ableitung des periodischen Systems* (Dresden: T. Steinkopff, 1910) and Morozov, *Die Offenbarung Johannis: eine astronomsich-historische Untersuchung* (Stuttgart, 1912).
12 Morozov, "Lingvističeskie spektry," 94.
13 Ibid.
14 Ibid.
15 Ibid., 101.
16 Ibid.
17 Ibid.
18 Ibid., 102.
19 Ibid., 100.
20 Ibid., 103
21 Ibid., 105.
22 Ibid., 109.
23 A. A. Markov, "Ob odnom primenenie statističeskogo metoda," *Izv. Imp. Akad. Nauk, Sankt Peterburg*, ser. 6, Vol. 10, no. 4 (1916), 239–42.
24 See George Kingsley Zipf, *Selected Studies of the Principle of Relative Frequency in Language* (Cambridge, MA: Harvard University Press, 1932); and Zipf, *The Psycho-Biology of Language: An Introduction to Dynamic Philology* (Cambridge, MA: Harvard University Press, 1935).
25 See also David Crystal, ed., *The Cambridge Encyclopedia of Language* (Cambridge: Cambridge University Press, 1987); and C. Prün, "G. K. Zipf's Conception of Language as an Early Prototype of Synergetic Linguistics," in *Journal of Quantitative Linguistics* 6 (1999).
26 George Kingsley Zipf, *Human Behavior and the Principle of Least Effort: An Introduction to Human Ecology* (Cambridge, MA: Harvard University Press, 1949), 24–35.
27 See Zipf, *Human Behavior and the Principle*, 33–35.
28 Mandelbrot, *The Fractal Geometry of Nature*, 422.
29 Benoît Mandelbrot, "Contribution à la théorie mathématique des jeux de communication" (PhD diss., l'Université de Paris, 1953), 2, 10.
30 A. Barcellos, "Interview of B. B. Mandelbrot," in *Mathematical People* (Boston: Birkhäuser, 1984), 3.
31 Claude E. Shannon and Warren Weaver, *The Mathematical Theory of Communication* (Urbana: University of Illinois Press, 2009), 10.
32 See M. Bullynck, "Markovketten—Variationen im Computer," in Hilgers and Velminski, eds., *Andrej A. Markov*, 162–64.
33 Bullynck, "Markovketten," 162–64.
34 Frieder Nake, *Ästhetik als Informationsverarbeitung. Grundlagen und Anwendungen der Informatik im Bereich ästhetischer Produktion und Kritik* (Vienna: Springer, 1974), 235.
35 Mandelbrot, *The Fractal Geometry of Nature*, 404.